ORBITAL MECHANICS OF PLANETS — GRADUATE EXAM

Total Points: None | Questions: 5 | Date: April 27, 2026

Diagram-based questions on Orbital Mechanics of Planets

Question 1

10.0 points

Analyze the diagram below showing the path of a planet orbiting a star. Use Kepler's second law to explain why the areas swept out by the line segment joining the planet and the star in equal time intervals are equal.

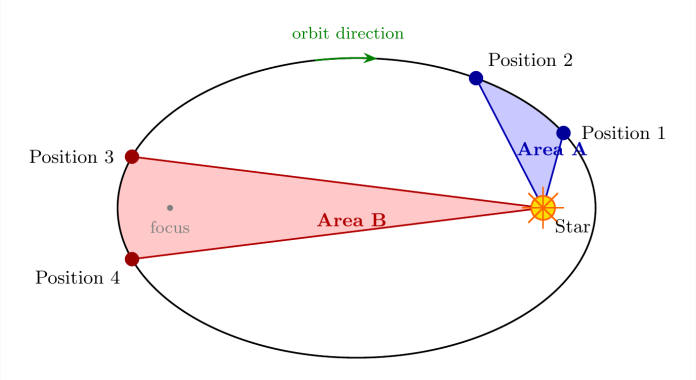

Fig. 1. Planetary Orbit

Question 2

10.0 points

Examine the diagram below of a Hohmann transfer orbit between two circular orbits. Describe the steps and velocities needed to perform this orbital transfer, including the initial, transfer, and final orbits.

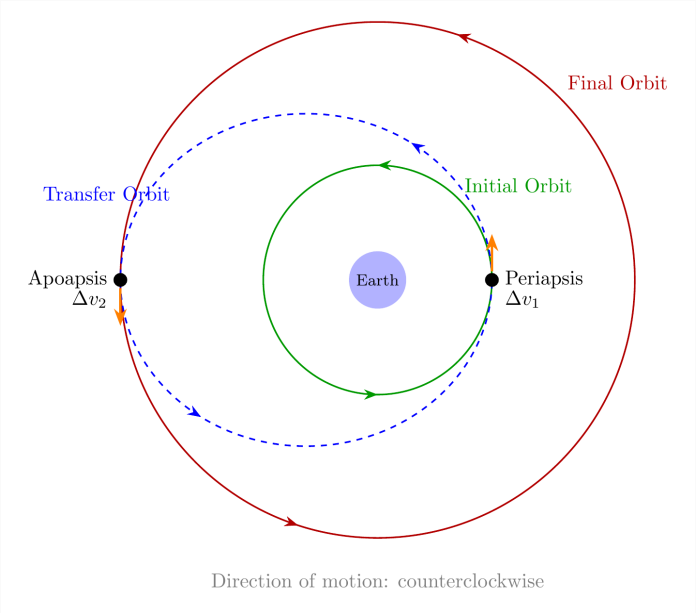

Fig. 2. Hohmann Transfer Orbit

Question 3

10.0 points

Analyze the diagram below depicting a gravitational slingshot maneuver (flyby) around a planet. Explain how the spacecraft's velocity is altered by this maneuver using principles of orbital mechanics.

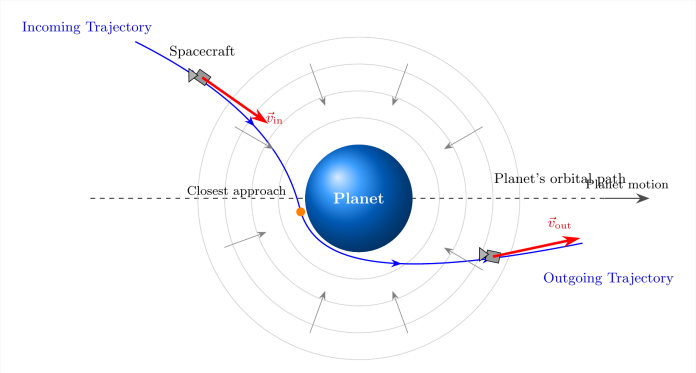

Fig. 3. Gravitational Slingshot

Question 4

10.0 points

Interpret the diagram below illustrating the Lagrange points (L1-L5) in the Earth-Sun system. Identify each point's location and provide a brief explanation of the gravitational and orbital dynamics that allow these points to exist.

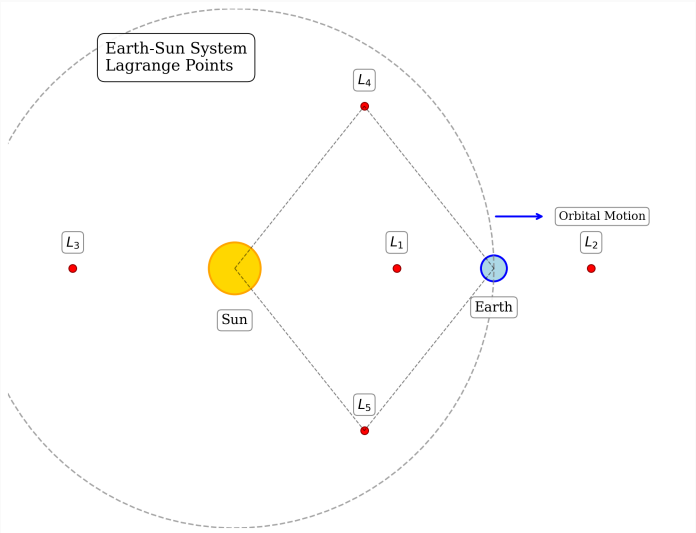

Fig. 4. Lagrange Points

Question 5 10.0 points

Examine the diagram below showing the precession of Mercury's orbit around the Sun. Explain the main factors that contribute to this precession, including relativistic effects and influences from other planets.

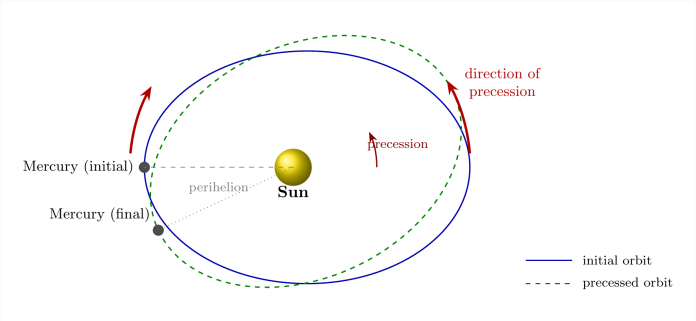

Fig. 5. Orbital Precession